TS1 is connected directly to the microcontroller. The touchscreen has four wires, namely, X+, X-, Y+ and Y-. All these wires are connected to four analogue-to-digital converter (ADC) pins of Port A of IC2. Wire X+ (pin 1) is connected to PA0, X- (pin 2) to PA1, Y+ (pin 3) to PA2 and Y- (pin 4) to PA3.

Data pins D0 through D7 of GLCD are connected to Port B pin PB0 through PB7 of ATmega16A, respectively. As 1MHz internal clock of ATmega16A is used, there is no need for an external crystal. RST pin 9 of IC2 is connected to ground through switch S1.

On pressing S1 momentarily, IC2 will get reset. Control pins (RS, R/$\overline{W}$, EN, etc) of the GLCD are connected to Port D of ATmega16A. There are two chips in GLCD that are controlled by control pins CS1 (pin 15) and CS2 (pin 16) connected to pins 17 and 18 of IC2, respectively. The reset pin (RST) of the GLCD (pin 17) is connected to pin 19 of IC2.

Since output of the touchscreen is analogue, the user needs to convert it into digital. This is done by the internal ADC of ATmega16A. Port A has an inbuilt ADC, so an external ADC chip (like ADC 0804) is not required.

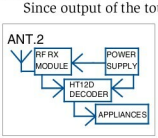

Fig. 3: Block diagram of receiver

Ports PC0 and PC1 of IC2 are connected to encoder IC3. All addresses from A0 to A7 of encoder IC3 and decoder IC5 are connected to ground in both transmitter and receiver circuits. For transmission, $\overline{TE}$ pin (pin 14) of IC3 is connected to ground. The output (DOUT) of IC3 is connected to data pin 2 of TX1 to transmit the bits through RF signals.

Working of the transmitter section is simple. When the user touches 'light on' symbol on the touchscreen, resistance of the touchscreen changes, and it changes the voltage across the touchscreen output. This voltage difference is converted to a digital signal by the ADC of ATmega16 microcontroller. ATmega16A checks the X and Y co-ordinates range. If it is in the predefined range, it sets the output according to the user's requirement at Port C. This signal is transmitted using 433MHz RF Tx module through an antenna (ANT.1).

Receiver section. Block diagram of the receiver section is shown in Fig. 3. The power supply unit is used to power the decoder and RF receiver module. The four-digit binary codes transmitted through the transmitter section are received by ANT.2 of the receiver circuit. Here, the decoder is used to decode the received signal. The microcontroller in the transmitter section switches the devices/appliances connected to the receiver section accordingly.

Circuit diagram of the receiver section is shown in Fig. 4. It is built

PARTS LIST

Semiconductors:
IC1, IC4 - 7805, 5V voltage regulator
IC2 - ATmega16A microcontroller
IC3 - HT12E encoder
IC5 - HT12D decoder
IC6 - ULN2003 relay driver
IC7 - 7812, 12V voltage regulator
LCD1 - 128X64 GLCD
BR1, BR2 - 1A bridge rectifier
LED1 - 5mm LED

Resistors (all 1/4-watt, ±5% carbon):
R1 - 100-ohm
R2 - 10-kilo-ohm
R3 - 1-mega-ohm
R4 - 33-kilo-ohm
R5 - 33-ohm
VR1 - 10-kilo-ohm potmeter

Capacitors:
C1, C2, C5-C7 - 0.1µF ceramic disk
C3, C4 - 1000µF, 25V electrolytic

Miscellaneous:
CON1 - 4-pin connector
CON2-CON4 - 2-pin terminal connector
S1 - Push-to-on switch
TS1 - 4-wire touchscreen
TX1 - 433MHz RF TX module
RX1 - 433MHz RF RX module
ANT.1, ANT.2 - 10cm copper wire
RL1, RL2 - 12V, one change-over relay
X1, X2 - 230V AC primary to 12V, 500mA secondary transformer
- 60W bulb
- Fan

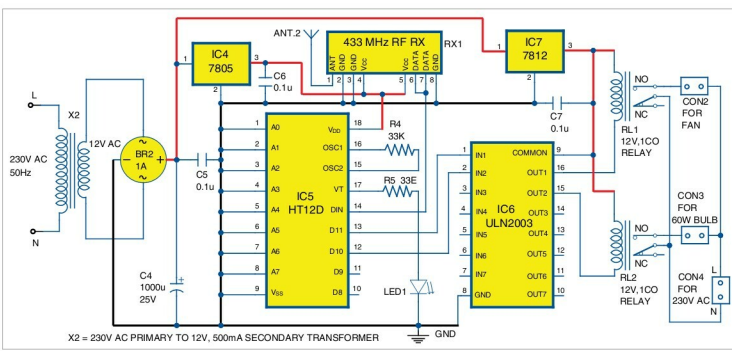

Fig. 4: Circuit diagram of the receiver